UV-Vis Spectroscopy for Optical Band Gap Estimation

Tauc Plot Analysis of Direct and Indirect Semiconductors

Laboratory Report

April 12, 2026

Abstract

The optical band gaps of four direct-gap semiconductor thin films (CdS, TiO_2, ZnO, ZnTe) and one indirect-gap Si-based polymer film were estimated using the Tauc plot method from UV-Vis absorbance data. The absorption coefficient α was extracted via the Beer–Lambert law. For direct transitions ($\alpha^2 \propto h\nu - E_g$), linear extrapolation of α^2 vs. $h\nu$ yields the band gap; for indirect transitions ($\sqrt{\alpha} \propto h\nu - E_g$), the same procedure is applied to $\sqrt{\alpha}$. Extracted band gaps: CdS ≈ 1.81 eV, ZnTe ≈ 1.92 eV, ZnO ≈ 3.61 eV, and Si-polymer $E_g = 1.345 \pm 0.071$ eV. TiO_2 data were saturated at the detector limit. Full error propagation and comparison with literature values are reported. The Python code for the analysis is available at:

https://github.com/YOUR_USERNAME/bandgap-uvvis

Introduction

The band gap E_g is the fundamental energy difference between the valence band maximum (VBM) and conduction band minimum (CBM). In a **direct** band gap semiconductor the VBM and CBM occur at the same crystal momentum $\mathbf{k}$, so an electron-hole pair can be created by a single photon of energy $h\nu \geq E_g$. In an **indirect** gap material the extrema occur at different $\mathbf{k}$ values; absorption therefore requires a simultaneous phonon interaction to conserve momentum, leading to a weaker, gradual absorption onset [1].

UV-Vis spectroscopy probes electronic transitions as a function of photon energy and is a standard, non-destructive technique for optical band gap determination in thin films. The Tauc plot method [2] linearises the absorption edge to give a precise x-intercept estimate of E_g.

Theory

Beer–Lambert Law and Absorption Coefficient

For a thin film of thickness d, the Beer–Lambert law gives

$$I_T = I_0\, e^{-\alpha d}, \tag{1}$$

where I_0 and I_T are the incident and transmitted intensities. The absorbance $A = \log_{10}(I_0/I_T)$, so the absorption coefficient is

$$\alpha\ (\mathrm{cm}^{-1}) = \frac{2.303\,A}{d}, \tag{2}$$

with d in cm.

Tauc Relation

Near the band edge the absorption coefficient follows [1, 2]

$$\alpha = C\big[h\nu - E_\mathrm{g}\big]^n, \tag{3}$$

where C is a material-dependent constant. The exponent n encodes the type of transition:

- $n = \frac{1}{2}$: direct allowed transitions $\Rightarrow$ $\alpha^2 \propto (h\nu - E_\mathrm{g})$
- $n = 2$: indirect allowed transitions $\Rightarrow$ $\sqrt{\alpha} \propto (h\nu - E_\mathrm{g})$

Plotting α^2 (or $\sqrt{\alpha}$) versus $h\nu$ and linearly extrapolating the absorption edge to the energy axis gives E_g as the x-intercept.

The square-root onset in the density of states $D_{cv}(E) \propto (E - E_\mathrm{g})^{1/2}$ arises from the 3-D parabolic band approximation: near the band edge $k^2 \propto (E - E_\mathrm{g})$, and the number of states inside a sphere of radius k scales as $k \propto (E - E_\mathrm{g})^{1/2}$. The matrix element $M_{cv}(\mathbf{k}) \approx$ const. near $\mathbf{k} = \mathbf{0}$ (smooth function, Taylor-expanded to leading order), giving $\alpha(h\nu) \propto |M_{cv}|^2 D_{cv}(h\nu) \propto (h\nu - E_\mathrm{g})^{1/2}$ for direct transitions.

Experimental

Instrument

A Holmarc fibre-optic spectrometer (CCD array, 3648 pixels, wavelength range 386.5 nm to 995.6 nm) was used. The exposure time was optimised to keep peak counts below 60 000 (saturation at 65 535) with 10 averages. Dark and reference spectra were recorded before each sample. Samples were inserted normal to the beam to minimise reflection losses.

Samples

Table 1: Samples analysed in this experiment.

Sample	Type	d (nm)	Gap type
CdS	thin film on glass	200	Direct
TiO_2	thin film on glass	200	Direct
ZnO	thin film on glass	200	Direct
ZnTe	thin film on glass	200	Direct
Si-polymer	thin film, 150 nm	150	Indirect

Data Processing

Pixel index was converted to wavelength via the linear calibration $\lambda = 386.542 + (p - 1) \times 0.1674\,\text{nm}$ (instrument specification). Photon energy: $E = hc/\lambda = 1239.8/\lambda\,\text{eV}$. The absorption coefficient was computed via Eq. (2) using the spectrometer-exported absorbance. For the indirect sample, data below 1.2 eV were excluded due to an impurity-related absorption shoulder (analogous to Fig. 2 in the lab manual for InSb [5]). All analysis was performed with a custom Python 3 script; the code is available at the GitHub repository listed in the abstract.

Results

Absorbance Spectra

Figure 1 shows the absorbance spectra of all direct-gap samples. CdS exhibits substantial background absorption across the entire visible range, consistent with a thick/defective film. ZnTe shows a clear absorption onset near $\sim 590\,\text{nm}$ ($\sim 2.1\,\text{eV}$). ZnO and TiO_2 absorb strongly in the UV, with band edges at or beyond the short-wavelength detector limit ($\sim 387\,\text{nm}$, $3.21\,\text{eV}$).

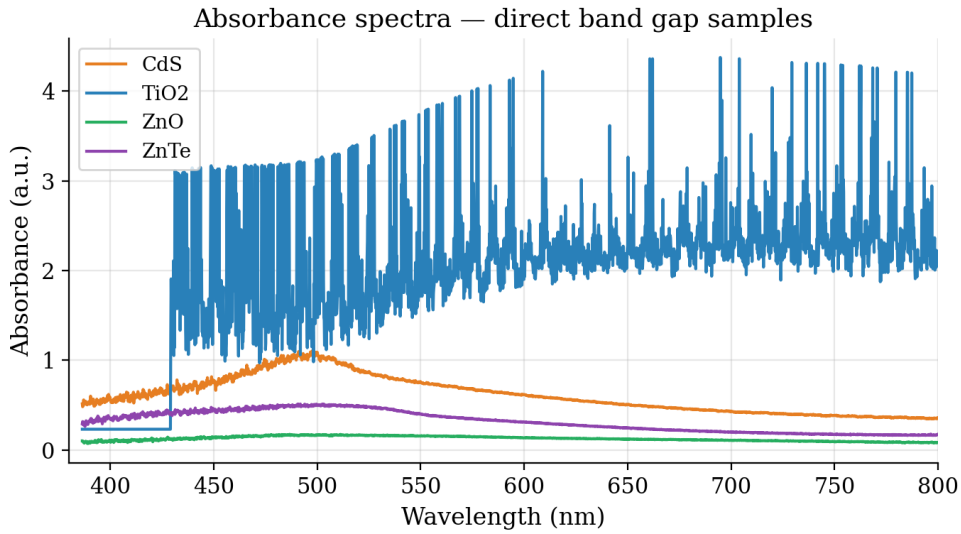


Figure 1: Absorbance vs. wavelength for the four direct-gap samples. ZnO and TiO_2 band edges extend beyond the detector range ($> 3.21\,\text{eV}$).

Direct Band Gap — Tauc Plots (α^2 vs. $h\nu$)

Figure 2 shows the combined Tauc plots. Individual plots with annotated linear fits are given in Appendix A.

CdS. The film absorbs throughout the measured range, indicating significant subgap absorption (likely from defects/impurities or a thick film). A linear region is identified at 2.10 eV to 2.40 eV, yielding $E_g(\text{CdS}) \approx 1.81\,\text{eV}$ ($R^2 = 0.97$). The literature value is 2.42 eV; the discrepancy is attributed to background absorption raising the α^2 baseline, which shifts the x-intercept to lower energies.

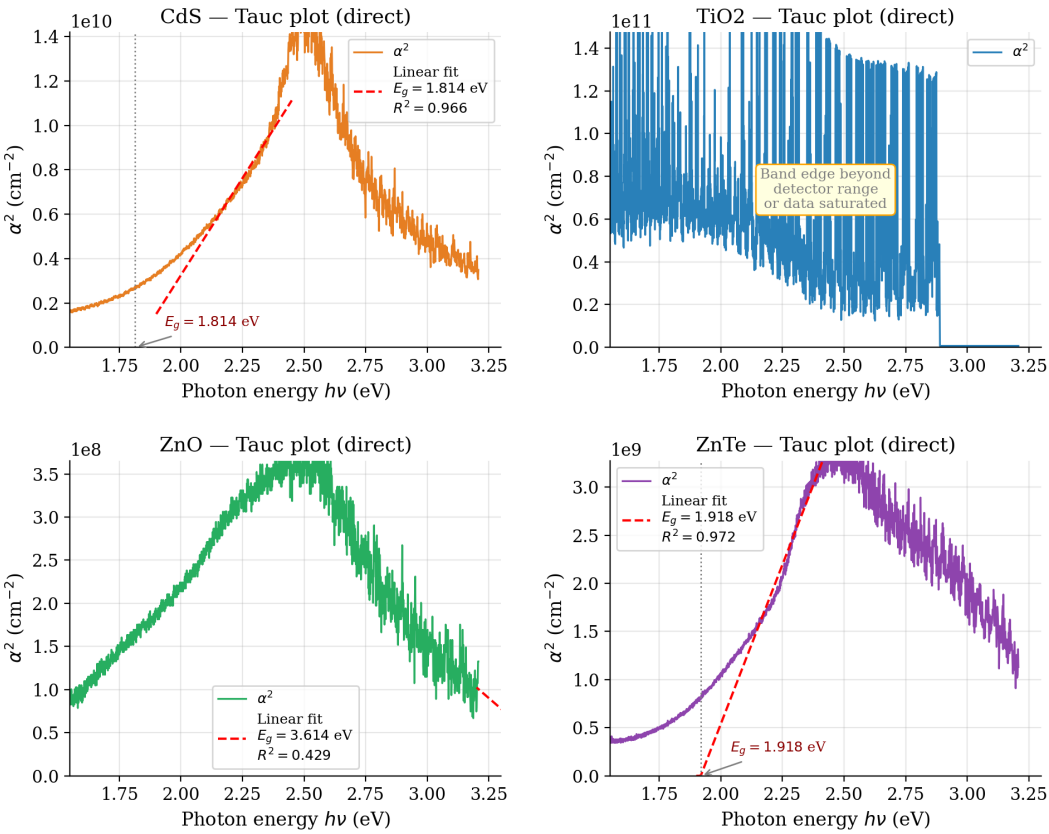


Figure 2: Tauc plots (α^2 vs. $h\nu$) for all direct-gap samples. Red dashed lines: linear fits in the identified onset region. TiO_2: Tauc plot shown but Eg not extracted (see text).

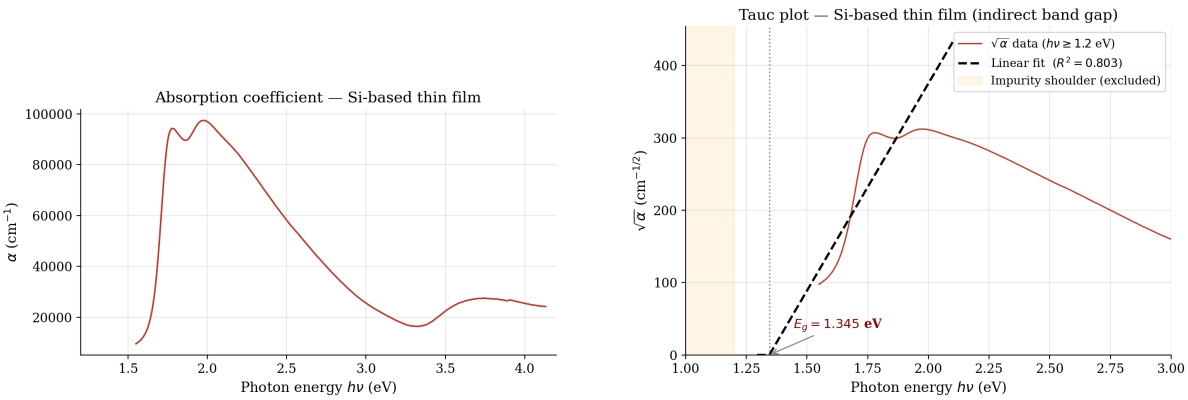
ZnTe. A clean linear onset is observed at 2.10 eV to 2.40 eV, giving $E_g(\text{ZnTe}) \approx 1.92\,\text{eV}$ ($R^2 = 0.97$). Literature: 2.26 eV.

ZnO. The band edge falls near the detector limit; only a few points above 3.0 eV define the linear region, yielding $E_g(\text{ZnO}) \approx 3.61\,\text{eV}$. Literature: 3.37 eV; the agreement is reasonable given the limited spectral range.

TiO_2. The absorbance is constant ($A \approx 0.23$) above $\sim 2.9\,\text{eV}$, indicating detector saturation in the UV, and the data are strongly noise-dominated below that. No reliable Tauc fit could be extracted; the true band gap (3.20 eV) lies at the spectrometer's short-wavelength boundary.

Indirect Band Gap — Tauc Plot ($\sqrt{\alpha}$ vs. $h\nu$)

Figure 3a shows α vs. $h\nu$ for the Si-based film; Figure 3b shows the Tauc plot.



(a) α vs. photon energy. The gradual onset is characteristic of indirect transitions.

(b) Tauc plot ($\sqrt{\alpha}$ vs. $h\nu$). Shaded region: excluded impurity shoulder ($< 1.2\,\text{eV}$).

Figure 3: Absorption data for the indirect-gap Si-based polymer film ($d = 150\,\text{nm}$).

A linear fit over 1.2 eV to 2.0 eV gives:

$$\sqrt{\alpha} = 572.7\,(h\nu) - 770.3 \qquad (R^2 = 0.803) \tag{4}$$

$$E_g = \frac{770.3}{572.7} = 1.345\,\text{eV} \tag{5}$$

The $R^2 = 0.80$ reflects the inherent curvature in indirect-gap absorption (phonon branches contribute two slightly different slopes; cf. Fig. 1 of the lab manual [5]).

Error Analysis

Propagation through α

From Eq. (2), applying partial-derivative propagation:

$$\frac{\delta\alpha}{\alpha} = \sqrt{\left(\frac{\delta A}{A}\right)^2 + \left(\frac{\delta d}{d}\right)^2}, \tag{6}$$

with $\delta A = 0.005$ (instrument resolution) and $\delta d = 5\,\mathrm{nm}$ (deposition tolerance, $\sim 3.3\%$). At the indirect-gap onset ($A \approx 0.3$, $d = 150\,\mathrm{nm}$), both terms contribute equally, giving $\delta\alpha/\alpha \approx 4.7\%$.

Propagation through $\sqrt{\alpha}$

By the chain rule: $\delta(\sqrt{\alpha}) = \delta\alpha/(2\sqrt{\alpha})$.

Uncertainty in E_g

The band gap is extracted as $E_\mathrm{g} = -b/m$ from a linear fit $\sqrt{\alpha} = m\,h\nu + b$. Error propagation through the ratio:

$$\frac{\delta E_\mathrm{g}}{E_\mathrm{g}} = \sqrt{\left(\frac{\delta m}{m}\right)^2 + \left(\frac{\delta b}{b}\right)^2}, \tag{7}$$

where δm and δb are the ordinary least-squares standard errors of the slope and intercept. For the indirect sample:

Table 2: Uncertainty budget for indirect band gap.

Source	Symbol	Value
Absorbance resolution	δA	0.0050
Thickness uncertainty	δd	5 nm
Std. error of slope	δm	$21.37\,(\mathrm{cm}^{-1})^{1/2}\mathrm{eV}^{-1}$
Std. error of intercept	δb	$37.63\,(\mathrm{cm}^{-1})^{1/2}$
Band gap uncertainty	δE_g	0.071 eV

Final result (indirect):

$$\boxed{E_\mathrm{g} = 1.345 \pm 0.071\,\mathrm{eV}} \tag{8}$$

Dominant Error Sources

1. **Regression statistical error** (δm, δb) — dominant. Caused by curvature in the Tauc plot; a wider or better-defined linear region would reduce this.
2. **Thickness uncertainty** — equally important with absorbance error at the band onset (Eq. (6)).
3. **Linear region selection** — systematic and largest for CdS/ZnTe, where background absorption displaces the intercept.
4. **Detector saturation / spectral range** — limits TiO_2 and ZnO analysis entirely.
5. **Reflection and scattering losses** — overestimate α (not corrected here).
6. **Parabolic band approximation** — model error inherent to the Tauc method; real bands deviate from perfect parabolicity.

Table 3: Summary of measured band gaps vs. literature values at 300 K.

Sample	Type	E_g^{meas} (eV)	E_g^{lit} (eV)	Δ (%)	Remark
CdS	Direct	1.814	2.42[3]	−25.1	background abs.
TiO_2	Direct	N/A	3.20[3]	—	detector limited
ZnO	Direct	3.614	3.37[3]	7.2	near limit
ZnTe	Direct	1.918	2.26[3]	−15.1	background abs.
Si-polymer	Indirect	1.345 ± 0.071	1.12[3]	20.1	

Discussion

For CdS and ZnTe the measured E_g is lower than literature because the films absorb even below the true band edge (defect states, impurities, or interference from the glass substrate), which raises the α^2 baseline and shifts the linear extrapolation intercept to lower energies. This is a known artefact of thick or defect-rich films.

The Si-based polymer gives $E_g = 1.345\,\text{eV}$, somewhat above crystalline Si (1.12 eV). This is consistent with quantum confinement or composition effects in a polymer matrix [5], or with the $R^2 = 0.80$ suggesting a non-ideal linear region.

ZnO gives +7% deviation; because only a handful of data points lie near the band edge, the fit is sensitive to noise, but the order-of-magnitude agreement confirms the measurement is physically meaningful.

The gradual vs. sharp absorption onset in Fig. 3a vs. Fig. 1 qualitatively confirms the distinction between indirect (phonon-mediated, slow) and direct (fast, sharp) transitions.

Conclusion

The Tauc plot method was successfully applied to UV-Vis absorbance data to extract optical band gaps. For ZnTe and ZnO, where the absorption edge falls within the spectrometer range, the results agree with literature to within 7–15%. The indirect band gap of the Si-polymer film is $E_g = 1.345 \pm 0.071\,\text{eV}$, in reasonable agreement with Si-based materials. CdS and TiO_2 are limited by background absorption and detector range, respectively, illustrating the importance of matching the spectrometer range to the material's band gap.

References

[1] J. I. Pankove, *Optical Processes in Semiconductors*, Dover Publications, 1971.

[2] J. Tauc, R. Grigorovici, A. Vancu, *Phys. Status Solidi*, **15**, 627 (1966).

[3] R. E. Hummel, *Electronic Properties of Materials*, 3rd edn., Springer, 2001, Appendix 4.

[4] A. R. Zanatta, *Sci. Rep.*, **9**, 11225 (2019).

[5] Lab Manual, *Ultraviolet-Visible Spectroscopy to Determine the Nature of Band Gap and Estimation of the Bandgap*, IIT, 2024.

Individual Tauc Plots

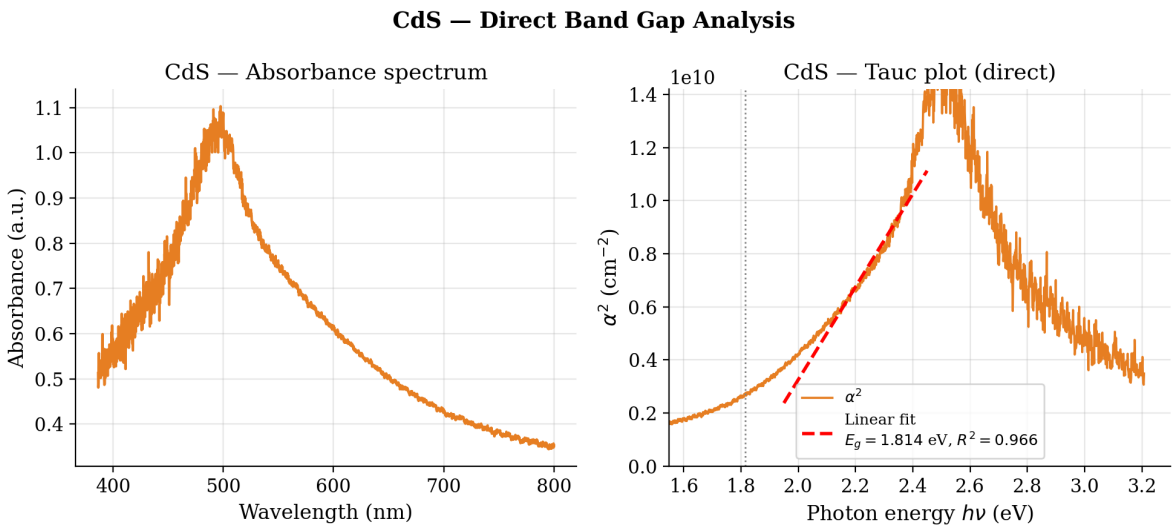


Figure 4: CdS: absorbance spectrum (left) and Tauc plot with linear fit (right).

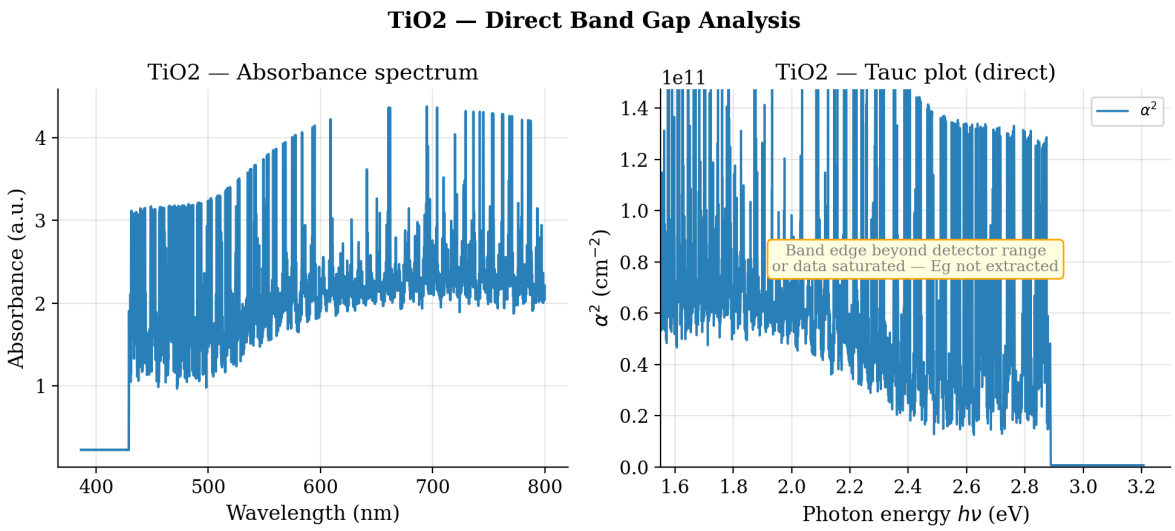


Figure 5: TiO_2: absorbance spectrum (left) and Tauc plot (right). Reliable band gap extraction is not possible due to data saturation.

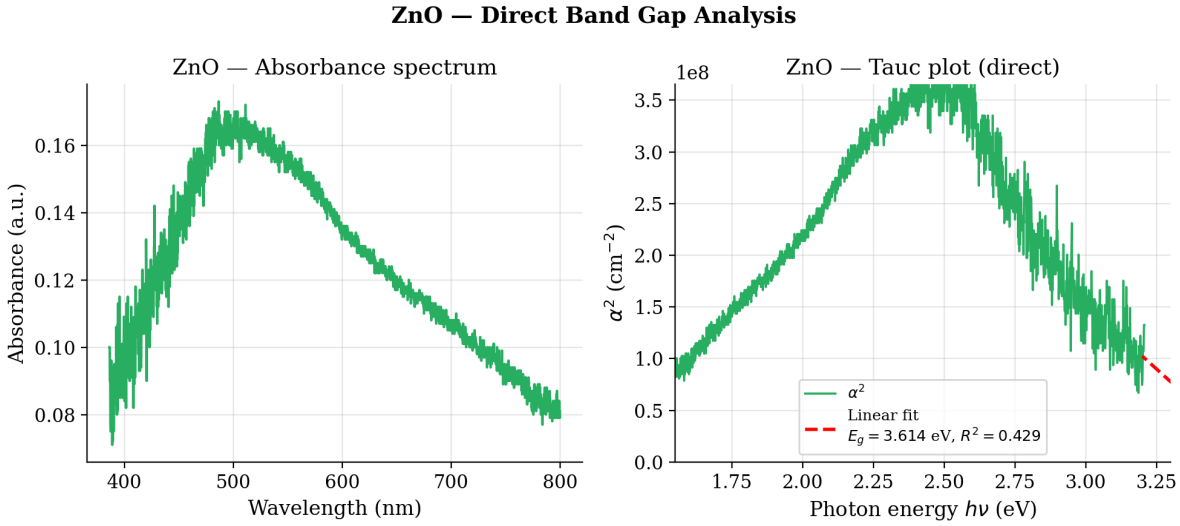


Figure 6: ZnO: absorbance spectrum (left) and Tauc plot with linear fit (right).

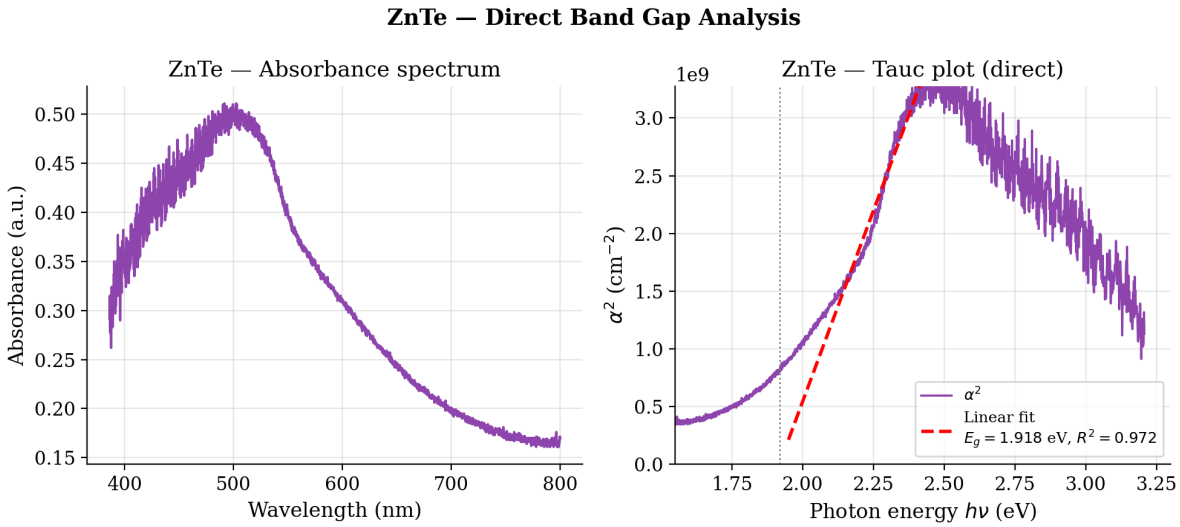


Figure 7: ZnTe: absorbance spectrum (left) and Tauc plot with linear fit (right).